

Fuzzy sets are used in artificial intelligence. Each element in the universal set U has a degree of membership, which is a real number between 0 and 1 (including 0 and 1), in a fuzzy set S . The fuzzy set S is denoted by listing the elements with their degrees of membership (elements with 0 degree of membership are not listed).

For instance, we write $\{0.6 \text{ Alice}, 0.9 \text{ Brian}, 0.4 \text{ Fred}, 0.1 \text{ Oscar}, 0.5 \text{ Rita}\}$ for the set F (of famous people) to indicate that Alice has a 0.6 degree of membership in F , Brian has a 0.9 degree of membership in F , Fred has a 0.4 degree of membership in F , Oscar has a 0.1 degree of membership in F , and Rita has a 0.5 degree of membership in F (so that Brian is the most famous and Oscar is the least famous of these people).

Also, suppose that R is the set of rich people with $R = \{0.4 \text{ Alice}, 0.8 \text{ Brian}, 0.2 \text{ Fred}, 0.9 \text{ Oscar}, 0.7 \text{ Rita}\}$.

73) The complement of a fuzzy set $\bar{S}$ is the set S , with the degree of the membership of an element in $\bar{S}$ equal to 1 minus the degree of membership of this element in S . Find $\bar{F}$ (the fuzzy set of people who are not famous) and $\bar{R}$ (the fuzzy set of people who are not rich).

Solution: $\bar{F} = \{0.4 \text{ Alice}, 0.1 \text{ Brian}, 0.6 \text{ Fred}, 0.9 \text{ Oscar}, 0.5 \text{ Rita}\}$
 $\bar{R} = \{0.6 \text{ Alice}, 0.2 \text{ Brian}, 0.8 \text{ Fred}, 0.1 \text{ Oscar}, 0.3 \text{ Rita}\}$

74) The union of two fuzzy sets S and T is the fuzzy set $S \cup T$, where the degree of membership of an element in $S \cup T$ is the maximum of the degrees of membership of this element in S and in T . Find the fuzzy set $F \cup R$ of rich or famous people.

Solution: $F \cup R = \{ \max(0.6, 0.4) \text{ Alice}, \max(0.9, 0.8) \text{ Brian}, \max(0.4, 0.2) \text{ Fred}, \max(0.1, 0.9) \text{ Oscar}, \max(0.5, 0.7) \text{ Rita} \}$
 $= \{0.6 \text{ Alice}, 0.9 \text{ Brian}, 0.4 \text{ Fred}, 0.9 \text{ Oscar}, 0.7 \text{ Rita}\}$

75) The intersection of two fuzzy sets S and T is the fuzzy set $S \cap T$, where the degree of membership of an element in $S \cap T$ is the minimum of the degrees of membership of this element in S and in T . Find the fuzzy set $F \cap R$ of rich and famous people.